1.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to Gray-scale.

1. Aim

To write and execute a Python program to read an image and convert it into grayscale using the OpenCV library.

2. Procedure

1. Import the OpenCV library using import cv2.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Convert the color image into grayscale using cv2.cvtColor().
5. Display the original image and grayscale image using cv2.imshow().
6. Press any key to close the image windows.

3. Code

```
import cv2


# Read the image
image = cv2.imread("img1.jpg")


# Check if image is loaded
if image is None:
    print("Image not found!")
else:
    print("Image read successfully")


    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


    # Display original image
    cv2.imshow("Original Image", image)


    # Display grayscale image
```

```
cv2.imshow("Grayscale Image", gray)


# Wait until a key is pressed
cv2.waitKey(0)


# Close all windows
cv2.destroyAllWindows()
```

4.Sample input:



5.sample output:



6. Result

Thus, the Python program to read an image and convert it into grayscale using the OpenCV library was executed successfully.